

Mechanical Robotic Gripper - Dimension and Calculation Sheet

Final SolidWorks design specifications

1. Final Design Dimensions

Component / Feature	Final Dimension / Location	Notes
Base plate	120 × 120 × 8 mm	Final trimmed base
Base main pivot holes	2 × Ø6.5 mm	
Pivot-hole spacing	80 mm center-to-center	
Left pivot center	20 mm from left, 80 mm from bottom	Coordinates: (20, 80) mm
Right pivot center	100 mm from left, 80 mm from bottom	Coordinates: (100, 80) mm
Center base cutout	50 mm wide × 70 mm tall	Open through top
Cutout horizontal location	Centered; X = 35 to 85 mm	
Cutout bottom	50 mm from bottom	
Cutout lower corner fillets	R5 mm	
Base outer fillets	R5 mm	As modeled
Slider slot	7 mm wide × 35 mm overall length	
Slot straight centerline length	28 mm	
Slot center	X = 60 mm, Y = 24.1 mm	On final 120 mm base
Finger	16 × 115 × 8 mm	2 required
Finger holes	2 × Ø6.5 mm per finger	
Finger hole centerline	8 mm from either side	Centered across 16 mm width
Lower finger/link hole	10 mm from bottom	
Main finger/base pivot	35 mm from bottom	
Finger hole spacing	25 mm center-to-center	
Connecting link	59 × 14 × 5 mm	2 required
Connecting-link holes	2 × Ø6.5 mm per link	
Connecting-link hole spacing	45 mm center-to-center	
Link hole centers from ends	7 mm	
Rounded link ends	R7 mm	
Center slider	40 × 30 × 8 mm	
Slider holes	4 × Ø6.5 mm	
Left linkage hole	(10, 15) mm	Coordinates from slider origin
Right linkage hole	(30, 15) mm	Coordinates from slider origin
Lower guide-pin hole	(20, 7.5) mm	Coordinates from slider origin
Upper guide-pin hole	(20, 22.5) mm	Coordinates from slider origin
Guide-hole spacing	15 mm	
Guide pins	2 × Ø6 × 16 mm	
Slider handle pin	1 × Ø6 × 16 mm	Manual push/pull input
Jaw pad	10 × 20 × 8 mm	2 required
Jaw gripping-end fillets	R2 mm	
Jaw grooves	3 grooves per jaw	
Groove size	6 × 1 mm	
Groove depth	0.75 mm	
Groove spacing	5 mm center-to-center	
Groove center heights	5, 10, and 15 mm from jaw bottom	
Linkage pivot pin	Ø6 × 13 mm shaft	4 required
Pivot-pin head	Ø10 × 2 mm	
Main finger bolts	2 × M6 × 25 mm, ISO 4017	
Main washers	2 × M6, ISO 7089	
Main nuts	2 × M6, ISO 4032	
Limit Angle mate	70° minimum to 90° maximum	Assembly constraint; not a manufactured dimension

2. Measured Performance

Parameter	Symbol	Value
Maximum jaw opening	Gmax	44.00 mm
Minimum jaw opening	Gmin	1.05 mm
Total jaw-gap change	ΔG	42.95 mm
Slider travel	Δx_s	9.14 mm
Handle travel	Δx_h	9.14 mm
Connecting-link pivot spacing	L	45 mm
Finger pivot-to-link-hole distance	r	25 mm
Base finger-pivot spacing	-	80 mm

3. Calculations

Calculation	Formula	Substitution	Result
Total jaw-opening change	$\Delta G = G_{\max} - G_{\min}$	44.00 - 1.05	42.95 mm
Travel of each jaw	$\Delta x_j = \Delta G / 2$	42.95 / 2	21.475 mm $\approx$ 21.48 mm
Average total jaw-motion ratio	$R = \Delta G / \Delta x_s$	42.95 / 9.14	4.70 : 1
Average individual-jaw motion ratio	$R_j = \Delta x_j / \Delta x_s$	21.475 / 9.14	2.35 : 1
Handle travel	$\Delta x_h = \Delta x_s$	9.14	9.14 mm
Measured jaw-opening range	Gmin to Gmax	1.05 to 44.00	1.05-44.00 mm

4. Calculation Notes

- The gripper is symmetric, so the total jaw-gap change is divided equally between the two jaws.
- The average total jaw-motion ratio means that 1 mm of slider/handle travel produces approximately 4.70 mm of total jaw-gap change over the measured range.
- Each individual jaw moves approximately 2.35 mm per 1 mm of slider/handle travel on average.
- The slider handle is rigidly attached to the center slider, so handle travel is equal to slider travel.
- The handle does not change the linkage geometry or the measured jaw-opening performance.
- The measured opening range describes the final CAD geometry; it does not by itself guarantee successful gripping of every object size within that range.

5. Final Calculation Summary

Item	Final Value
Maximum jaw opening	44.00 mm
Minimum jaw opening	1.05 mm
Total jaw-gap change	42.95 mm
Slider/handle travel	9.14 mm
Travel per individual jaw	21.48 mm
Average total jaw-motion ratio	4.70 : 1
Average individual-jaw motion ratio	2.35 : 1